

Table of Physical Constants

Electron rest mass	m_e	$9.109 \times 10^{-31} \text{ kg}$
Proton rest mass	M_p	$1.6726 \times 10^{-27} \text{ kg}$
Electronic charge	e	$1.6022 \times 10^{-19} \text{ C}$
Speed of light in free space	c	$2.9979 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ J s}$
Reduced Planck's constant	$\hbar = h/2\pi$	$1.0546 \times 10^{-34} \text{ J s}$
	$\hbar c$	197.33 MeV fm
Boltzmann's constant	k_B	$1.3807 \times 10^{-23} \text{ J K}^{-1}$
Gas constant	$\mathcal{R} = k_B/m_H$	$8.250 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Molar gas constant	R	$8.315 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Standard molar volume		$22.414 \times 10^{-3} \text{ m}^3 \text{ mol}^{-1}$
Unified atomic mass unit (^{12}C scale)	u	$931.5 \text{ MeV}/c^2 = 1.660538 \times 10^{-27} \text{ kg}$
Mass of hydrogen atom	m_H	$1.0078u = 1.6735 \times 10^{-27} \text{ kg}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 \text{ or J T}^{-1}$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 \text{ or J T}^{-1}$
Proton magnetic moment	μ_p	$2.7928\mu_N$
Neutron magnetic moment	μ_n	$-1.9130\mu_N$
Bohr radius	a_0	$5.292 \times 10^{-11} \text{ m}$
Fine structure constant	$\alpha = e^2/(4\pi\epsilon_0\hbar c)$	$(137.04)^{-1}$
Compton wavelength of electron	$\lambda_C = h/(m_e c)$	$2.4263 \times 10^{-12} \text{ m}$
Rydberg's constant	R_∞	$1.0974 \times 10^7 \text{ m}^{-1}$
	$R_\infty \hbar c$	13.606 eV
Stefan-Boltzmann constant	σ	$5.671 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Radiation density constant	$a = 4\sigma/c$	$7.561 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$
Gravitational constant	G	$6.673 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

Rest masses of some leptons and hadrons in MeV/c^2 :

$e^\pm$ 0.5110, $\mu^\pm$ 105.66, $\tau^\pm$ 1777, π^0 134.98, $\pi^\pm$ 139.57, $K^\pm$ 493.7, K^0 497.7, η 547, D^0 1865, $D^\pm$ 1869, p 938.3, n 939.6, Λ^0 1115.7, Σ^+ 1189, Σ^0 1193, Σ^- 1197, Ξ^0 1315, Ξ^- 1321, Ω^- 1672, Z^0 91.187×10^3 , $W^\pm$ 80.41×10^3 .

Quark	Charge	I_3	S	C	B	T	Mass (GeV/c^2)
u	$+\frac{2}{3}$	$\frac{1}{2}$	0	0	0	0	~ 0.003
d	$-\frac{1}{3}$	$-\frac{1}{2}$	0	0	0	0	~ 0.006
c	$+\frac{2}{3}$	0	0	+1	0	0	~ 1.25
s	$-\frac{1}{3}$	0	-1	0	0	0	~ 0.11
t	$+\frac{2}{3}$	0	0	0	0	+1	174.3
b	$-\frac{1}{3}$	0	0	0	-1	0	4.2

Astrophysical Data

1 astronomical unit	AU	$1.496 \times 10^{11} \text{ m}$
1 parsec	pc	$3.086 \times 10^{16} \text{ m}$
Luminosity of Sun	$L_{\odot}$	$3.85 \times 10^{26} \text{ W}$
Mass of Sun	$M_{\odot}$	$1.989 \times 10^{30} \text{ kg}$
Radius of Sun	$R_{\odot}$	$6.96 \times 10^8 \text{ m}$
Mass of Earth	M_E	$5.9742 \times 10^{24} \text{ kg}$
Radius of Earth	R_E	$6.3781 \times 10^6 \text{ m}$

Other data and conversion factors

1 ångstrom	Å	10^{-10} m
1 fermi	fm	10^{-15} m
1 barn	b	10^{-28} m^2
1 pascal	Pa	1 Nm^{-2}
1 standard atmosphere		$1.0132 \times 10^5 \text{ Pa}$
Standard acceleration due to gravity	g	9.807 m s^{-2}
1 electron volt	eV	$1.6022 \times 10^{-19} \text{ J}$
	eV/ hc	$8.065 \times 10^5 \text{ m}^{-1}$
	eV/ k_B	$1.1604 \times 10^4 \text{ K}$
Wavelength of 1 eV photon		$1.2399 \times 10^{-6} \text{ m}$

Trigonometrical identities

$$\sin(\theta + \phi) = \sin(\theta) \cos(\phi) + \cos(\theta) \sin(\phi)$$

$$\cos(\theta + \phi) = \cos(\theta) \cos(\phi) - \sin(\theta) \sin(\phi)$$

$$\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

$$\cos \alpha - \cos \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\beta - \alpha)$$

$$\text{In a triangle ABC, } a/\sin A = b/\sin B = c/\sin C$$

$$\text{and } a^2 = b^2 + c^2 - 2bc \cos A$$

Prefixes

$$\text{T} = \text{tera} = 10^{12}$$

$$\text{G} = \text{giga} = 10^9$$

$$\text{M} = \text{mega} = 10^6$$

$$\text{k} = \text{kilo} = 10^3$$

$$\text{c} = \text{centi} = 10^{-2}$$

$$\text{m} = \text{milli} = 10^{-3}$$

$$\mu = \text{micro} = 10^{-6}$$

$$\text{n} = \text{nano} = 10^{-9}$$

$$\text{p} = \text{pico} = 10^{-12}$$

$$\text{f} = \text{femto} = 10^{-15}$$